REPORT

Problem Description

Waste management is an important environmental and public health concern in Sierra Leone. In some communities, the collection and disposal of household waste can be inconsistent, resulting in waste accumulating along roadsides, in open spaces, and around residential areas. Poor waste management can contribute to environmental pollution, unpleasant surroundings, blocked drainage systems, and increased health risks.

One challenge is the difficulty residents may experience when reporting uncollected waste and obtaining information about collection requests. Without an organized reporting and tracking process, waste collection officers may struggle to prioritize requests, monitor outstanding complaints, and keep residents informed.

The proposed Waste Collection Management System aims to address this problem by providing a simple, structured way to register waste collection requests, record relevant information, monitor request statuses, and support the organization of waste collection activities.

Background

Waste collection requires coordination between residents, waste collection workers, and the authorities responsible for maintaining clean communities. An effective process should allow residents to report waste collection needs and provide responsible officers with the information required to manage those requests.

Where reporting relies mainly on informal communication or unorganized records, requests may be overlooked, duplicated, or difficult to track. A basic computerized system can help organize these activities by storing request details in a consistent format and making their status easier to monitor.

The proposed Waste Collection Management System is designed as a logical solution to this challenge. It will use basic programming logic concepts, including input, processing, output, conditional statements, repetition, and modular design. The project will demonstrate how these concepts can be applied to a practical community problem without requiring a complex or expensive technological infrastructure.

Community Impact

The proposed system could provide several benefits to residents and waste collection officers.

First, it would give residents a structured method of submitting waste collection requests. Each request could contain details such as the resident's name, location, type of waste, and date of submission.

Second, the system would help collection officers organize and monitor reported requests. By maintaining request records and updating their statuses, officers could identify requests that are pending, in progress, or completed.

Third, an organized record of requests could support better planning and help officers identify locations where waste collection services are frequently needed. Improved coordination could contribute to cleaner surroundings and greater community awareness of proper waste management.

However, the system would not automatically remove waste or guarantee that every request would be addressed immediately. Its effectiveness would depend on the availability of collection workers, transport, equipment, and appropriate waste disposal facilities.

Proposed Solution

The proposed solution is a Waste Collection Management System that records, processes, and tracks waste collection requests. It is intended to provide a simple and organized approach to communication between residents and waste collection officers.

The system will allow a resident to submit a request by providing the required information. It will validate the input to ensure that essential details are supplied. If the information is incomplete or invalid, the system will display an appropriate message and request correction. If the information is valid, the system will register the request and assign it a unique identification number.

The system will store the request with an initial status of "Pending." An authorized collection officer will be able to view registered requests and update their status to "In Progress" or "Completed" as work proceeds. Residents or responsible officers will be able to check a request's status using its identification number.

The main functions of the system will include request registration, input validation, request identification, request-status tracking, and the display of request information. These functions will be designed using algorithms, pseudocode, flowcharts, and modular programming logic.

Computational Thinking Process

Computational thinking involves breaking a problem into smaller parts and developing a logical solution that can be understood and implemented systematically. Four important aspects of computational thinking will be applied to the proposed system.

1. **Decomposition:** Decomposition involves dividing a complex problem into smaller, manageable tasks. The waste collection problem will be divided into request registration, input validation, request storage, request-status updating, and request-status checking. Each task will have a defined responsibility, making the overall system easier to design and understand.

2. **Pattern Recognition:** Pattern recognition involves identifying similarities and repeated situations within a problem. In the proposed system, every waste collection request will require similar information, including a location, waste description, submission date, and status. Requests will also follow a common process from submission to completion. Recognizing these patterns will help establish consistent rules for handling requests.
3. **Abstraction:** Abstraction involves concentrating on the information necessary to solve a problem while excluding unnecessary details. The system will focus on essential request information, including the request identification number, resident details, location, waste description, submission date, and status. Unrelated information will be excluded from the initial design to keep the system simple and manageable.
4. **Algorithm Design:** Algorithm design involves arranging instructions in a logical sequence to solve a problem. In this project, the algorithm will guide the registration, validation, storage, and tracking of waste collection requests. Conditional statements will determine whether submitted information is valid, while repetition may be used to process multiple requests. Functions and modules will organize related operations into separate components.

Together, these computational thinking techniques will help transform the waste collection problem into a structured solution that can be represented using an algorithm, pseudocode, and a flowchart.

Algorithm Design Approach

The system will follow a sequence of logical steps. It will begin by presenting the available operations to the user, such as registering a new request or checking an existing request. When a new request is submitted, the system will collect the required details and validate the information.

If the information is incomplete, the system will display an error message and allow the user to try again. If the information is valid, the system will generate a unique request identification number, save the request, assign it a pending status, and display a confirmation message.

When a request-status check is selected, the system will ask for the identification number and search the stored records. If a matching request exists, its current status will be displayed. Otherwise, the system will inform the user that the request could not be found.

The system will continue accepting operations until the user chooses to exit. This approach will provide a clear foundation for the detailed algorithm, pseudocode, flowchart, and functional modules presented in the following sections.

Sustainable Development Goal (SDG) Alignment

The proposed Waste Collection Management System supports Sustainable Development Goal 11 (SDG 11): Sustainable Cities and Communities. This goal includes making human settlements more inclusive, safe, resilient, and sustainable. Organizing waste collection requests can contribute to cleaner communities and better coordination of local waste management activities.

The project is also relevant to SDG 12: Responsible Consumption and Production, particularly the broader objective of improving waste management and reducing environmental impacts. By making collection requests easier to record and monitor, the proposed system could support more responsible waste handling.

The system would contribute to these goals indirectly. Its actual impact would depend on whether the information recorded leads to effective waste collection, appropriate disposal, and sustained participation by residents and responsible authorities.

Open-Source Justification

The proposed system would be suitable for development as open-source software. Open-source software makes its source code available under a license that permits users to inspect, use, modify, and redistribute it according to the license conditions.

Making the Waste Collection Management System open source could allow students, developers, community organizations, and local authorities to examine its design and adapt it to their needs. For example, another community could modify the system to include additional collection locations, different waste categories, or locally appropriate request procedures.

Open-source development could also support learning, collaboration, and improvement by allowing contributors to identify errors and suggest enhancements. A public repository could provide access to the project's documentation, algorithms, pseudocode, and future implementation.

Nevertheless, making the source code publicly available would not automatically provide the funding, equipment, personnel, or infrastructure needed to operate a real waste collection service. The system would still require responsible maintenance, appropriate data protection, and support from relevant stakeholders.

Digital Public Goods (DPG) Justification

Digital Public Goods are digital solutions that can provide public benefit and meet relevant standards for openness, privacy, security, and responsible use. Open-source software can be a foundation for a Digital Public Good, although being open source alone does not automatically qualify a project as one.

The proposed Waste Collection Management System has potential public value because it addresses a community service challenge that affects environmental cleanliness and public well-being. If developed and maintained responsibly, it could be adapted for use

by different communities and local authorities without requiring each community to design a completely new system.

To strengthen its potential as a Digital Public Good, the system would need clear documentation, an appropriate open-source license, accessible design, secure handling of resident information, and a sustainable maintenance plan. Its development should also consider communities with limited internet access and limited technological resources.

The project therefore represents an educational example of how a simple digital solution could support a public service. Its suitability as a Digital Public Good would depend on meeting the relevant requirements and demonstrating practical value to the communities it serves.